

1. (2 pts) If A is a 3×5 matrix and B is a 5×2 matrix, what is the size of the matrix product AB?

$$3 \times 2$$

2. (2 pts) If A is a 3×5 matrix and the product AB is a 3×5 matrix, what must be the size of the matrix B?

$$5 \times 5$$

3. (4 pts) If $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ compute the matrix products shown below.

a) $\mathbf{u}^T \mathbf{v} =$

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = [a + 2b + 3c]$$

b) $\mathbf{u} \mathbf{v}^T = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} a & b & c \\ 2a & 2b & 2c \\ 3a & 3b & 3c \end{bmatrix}$

4. (4 pts) Complete the following sentence.

a) What is the determinant of the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$? $\det A = ad - bc$

b) How is the determinant of A used to find A^{-1} ?

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

5. (3 pts) What can you say about the solution to the matrix equation $A\mathbf{x} = \mathbf{b}$ if A is an invertible matrix?

$A\vec{x} = \vec{b}$ will have a unique solution

$$\mathbf{x} = A^{-1} \vec{b}$$